Raj Ranjit Yadav

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Summary

MS Computer Science student at Northeastern University with one year as a Software Engineer at LTM, building C#/.NET backend features, REST APIs, SQL data work, and Azure DevOps CI/CD, and a Python PDF comparison tool that cut 50-page document validation from 5–6 hours to about 10 minutes (97%). Builds applied AI: governed multi-agent AI on AWS (Bedrock AgentCore, Lambda, Cedar), semantic search with pgvector and a local Llama 3.2 LLM, and a TensorFlow/Keras NLP classifier. Seeking software, AI/ML, or data co-ops and internships from May 2027.

Education

Northeastern University — Seattle, WA
Master of Science in Computer Science, GPA: 4.0/4.0 — *Expected May 2028*

- **Relevant Coursework:** Programming Design Paradigms, Database Management Systems

Nutan College of Engineering and Research — Pune, India
Bachelor of Technology in Computer Science and Engineering, CGPA: 7.58/10 — *Jul. 2025*

- **Relevant Coursework:** Data Structures, Design & Analysis of Algorithms, Database Systems, Operating Systems, Machine Learning, Artificial Intelligence, Cloud Computing, Big Data Analytics

Experience

LTM — Pune, India
Software Engineer — *Sep. 2025 – Aug. 2026*

- Migrated 300+ customer-facing insurance documents from a legacy DocMan platform to SmartComm as part of a four-member backend engineering team, covering quote, binding, and policy correspondence.
- Developed a Python PDF comparison tool that automatically flags differences between contract document versions, reducing comparison time for 50-page documents from 5–6 hours to approximately 10 minutes (97% reduction).
- Investigated legacy document-generation workflows and mapped and validated 1,000+ attributes across questionnaire data, template configurations, and backend data models, diagnosing data and output discrepancies.
- Compared new and existing SmartComm contract templates to track field additions, removals, and changes, performing impact analysis and giving development and testing teams mapping documentation.
- Developed and maintained backend functionality using C#, .NET Framework, and REST APIs, troubleshooting defects and supporting delivery of migration requirements ahead of deadlines.
- Executed SQL queries against a cloud-hosted database to retrieve insurance templates for quote letters and other correspondence, supporting template selection and dynamic data population.
- Collaborated in Agile/Scrum workflows with daily code reviews and technical documentation, using Azure DevOps CI/CD pipelines for automated build, validation, and deployment.

Jaguar Land Rover India Ltd. — Mumbai, India
IT Intern – IT Department — *Jul. 2024 – Sep. 2024*

- Automated daily data collection by building Python RPA scripts and Microsoft Power Automate flows that logged into internal dashboards, gathered daily data, and compiled it into Excel workbooks for downstream analysis.
- Consolidated multiple Python RPA processes to run concurrently on a single device instead of separate machines.
- Automated retrieval and processing of email data, replacing manual steps in recurring IT workflows.
- Built modular, reusable automation components with input validation, exception handling, and workflow testing.
- Documented automation workflows and supported teammates during deployment and troubleshooting.

Technical Skills

Languages: Python, C#, SQL, Java, JavaScript
Backend & Web: .NET Framework, .NET Web API, REST APIs, Entity Framework Core, LINQ, FastAPI, React.js
AI / ML: TensorFlow/Keras, NLP, LSTM, embeddings, semantic search, LLMs, RAG, multi-agent systems
Data: ETL pipelines, pandas, NumPy, Matplotlib, data validation, schema design, Excel, Streamlit, Gradio
Databases: PostgreSQL, pgvector, Microsoft SQL Server, MySQL, Amazon RDS, DynamoDB, Firebase, Qdrant, Neo4j
Cloud & DevOps: AWS (EC2, Lambda, S3, RDS, Bedrock, IAM), Azure DevOps, CI/CD, Git, Docker
Practices & Tools: OOP, data structures & algorithms, Agile/Scrum, SmartComm, impact analysis, Power Automate

Projects

Pre-Mortem – Governed Multi-Agent AI | *Python, AWS Lambda, Bedrock AgentCore, Cedar* Nov. 2026

- Built, as a team at the Breaking Barriers @ FYUZ 2026 hackathon (Track 3), triage, change, and report agents (Strands on Amazon Bedrock AgentCore) that diagnose network faults and propose safe fixes.
- Designed a 7-step AWS Lambda checkpoint for every model, tool, and agent call: registry, Bedrock Guardrails, digital-twin prediction, risk score, deny-by-default Cedar policies, human approval, and audit by trace ID.
- Applied least-privilege IAM so agents can invoke only the checkpoint and their own model; withdrawing a model in the registry pauses every call to it and escalates to a human.
- Turned regulator reports on the Rogers (2022), Optus (2023), and AT&T (2024) outages into Cedar rules and replay tests on a synthetic network, plus about 60 attack and benign scenarios frozen and hashed before runs.

Job Match Assistant | *Python, PostgreSQL, pgvector, Docker, pandas, Sentence Transformers, Ollama, Streamlit*

- Built an automated ETL pipeline ingesting 750+ job postings from a REST API with pagination and exponential-backoff retries, deduplicating data into a normalized PostgreSQL schema with idempotent upserts.
- Developed semantic job matching with sentence-transformer embeddings and pgvector cosine similarity over chunked resume embeddings, to rank 700+ job postings against a resume.
- Integrated a locally hosted Llama 3.2 model through Ollama to generate job-specific resume bullets, using prompt constraints designed to prevent fabricated experience, alongside per-job skill-gap analysis.
- Built a Streamlit dashboard to visualize skill demand, hiring locations, and salary trends across collected postings.

Toxic Comment Classifier | *Python, TensorFlow/Keras, NLP, NumPy, pandas, Matplotlib, Gradio*

- Trained a TensorFlow/Keras Bidirectional LSTM multi-label classifier on the Jigsaw Toxic Comment dataset, predicting six toxicity categories through independent sigmoid outputs with binary cross-entropy.
- Converted raw comments into integer sequences with a Keras TextVectorization layer over a 200,000-token vocabulary, feeding a learned 32-dimensional embedding layer and stacked dense layers.
- Deployed the trained model through an interactive Gradio web interface that scores comments in six categories.

Smart Task Management System | *C#, .NET Web API, Entity Framework Core, React, SQL*

- Engineered a full-stack task management platform with RESTful CRUD APIs using .NET Web API, Entity Framework Core, React, and SQL, applying OOP, the Repository Pattern, LINQ, and async programming.
- Implemented JWT authentication and role-based authorization to secure APIs and control access to resources.

Certifications

Microsoft Certified: Azure AI Fundamentals (AI-900) Feb. 2026
Microsoft Certified: Azure Fundamentals (AZ-900) Jan. 2026

Publication

IoT Based Automatic Accident Detection & Rescue System Using GPS – co-author, ICHISS-2024 Dec. 2024

Leadership & Activities

Computer Engineering Students' Association (CESA) Pune, India
Co-Treasurer and Event Head, Automobile Expo *Aug. 2023 – Jan. 2025*

- Led 30+ volunteers and coordinated an Automobile Expo attended by 3,000+ participants.
- Managed budgeting, vendor coordination, and financial planning for student activities as Co-Treasurer.
- Helped professors organize and run a college hackathon, coordinating the event for student participants.

College Cricket Team | *Team Captain*

- Led the college team in inter-college tournaments, coordinating team strategy and player participation.